STAFF REVIEW · H446 078 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Sorted data.

2. First possible candidate index.

3. Index3.

4. mid3=7, mid5=11, mid6=13, mid7=15; found index7.

I Do — source worked steps

1. Check the working — mid3=11, mid5=17, mid4=14, returns index4.

We Do — guided CHECK

1. Check the working
mid2=5, mid0=1, mid1=3, low2>high1: return−1.

You Do — source mark schemes

1. mid3=7 [2], low4 [1], mid5=11 [2], return5 [1].

2. equal return mid [2], smaller array mid→low=mid+1 [2], larger→high=mid−1 [2].

3. Linear O(n) [2] unsorted okay [1]; binary O(log n) [2] requires sorted data [1].

Exit Ticket — source answer

1. False. Use mid+1.